

Question for a Uni Design Project: Is the massive energy footprint of AI actually on your radar?

Posted by u/TalNix77 • 0 points • 8 comments

Hi everyone,

I'm a design student researching the "invisible" energy consumption of AI for a university project.

While the utility of tools like ChatGPT is obvious, the physical resources required to run them are massive. Studies suggest that a single generative AI query can consume significantly more energy than a standard web search (some estimates range from 10x to 25x more).

I'm looking for honest perspectives on this:

1. Awareness: Before reading this, were you actually aware of the scale of energy difference between a standard search and an AI prompt? Or is that completely "invisible" in your daily usage?
2. Impact on Usage: Does the energy intensity play any role in how you use these tools? Or is the utility simply the only factor that matters for your workflow?
3. Value vs. Waste: Do you view this high energy consumption as a fair investment for the results you get, or does the current technology feel inefficient to you?

I'm trying to get a realistic picture of whether this topic actually plays a role in users' minds or if performance is the priority.

Thanks for your input. <3

Comments

u/CyberGuyFlying • 1 points • 2025-12-08 08:03 UTC

I was tracking that training AI requires massive amounts of energy for data centers, which is a significant issue. But for each individual query, I was tracking that ever since the DeepSeek release,

more efficient AI technologies have emerged to make individual queries less demanding.

The AI query costing its parent company 0.02 cents or whatever instead of my Google search costing Google 0.001 cents is not on my radar whatsoever. Factors that are more important to me include: Speed, ease of use, quality of answer, etc.

u/BadGrampy • 2 points • 2025-12-07 06:05 UTC

If every query uses 25 times the energy of a regular search and I do an average of 20/day, that's 600/month, so 15000 regular searches. Multiply that by the billions of users. Google says each search costs around 0.0003 kWh/search. $.0003 \times 600 = .18\text{kWh}$. $.18\text{kWh} \times 25 = 4.5\text{kWh}$. $4.5\text{kWh} \times \$1.807 = \8.1315

As close as I could find, average usage in usa per person is 370kwh/month. Add 4.5wh. 374.5kwh/month.

1.2% increase.

Is this a massive energy footprint?

u/[deleted] • 1 points • 2025-12-06 03:22 UTC

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u/graymalkcat • 3 points • 2025-12-06 00:24 UTC

I ignore stuff like this because it's designed to drive some kind of anti-tech narrative.

u/Federal_Decision_608 • 1 points • 2025-12-05 23:48 UTC

Luddite bullshite

u/brainspark10-4 • 3 points • 2025-12-05 23:42 UTC

I read this and don't worry about it: <https://andymasley.substack.com/p/a-cheat-sheet-for-conversations-about>

u/Educational_Bar2807 • 0 points • 2025-12-05 23:38 UTC

They should be required to put little meters in the interface so people can see how much energy they use in a day. But no one will care till energy costs skyrocket and some people won't be able to afford to power their home anymore.

u/FIREaus67 • -1 points • 2025-12-05 21:34 UTC

Cudos to you for picking this topic. It's the ugly truth of AI that we keep locked up in the basement.

1. I hadn't seen that stat before. But was aware of the massive energy consumption of AI generally.

2. No it doesn't (and it should).

3. It feels to me like AI is far from efficient. We spend so much time correcting errors, hallucinations and misses. Only to have the same mistakes repeated. Simple example - was using Nano Banana yesterday to put an 80 year old migrant in an image of his home town in Greece. No matter the prompt, Gemini just kept repeating the same crappy image over and over. That morning it had done a brilliant job with a similar request - without error. What percentage of AI usage is due to AI failure? In my limited experience, I'd estimate around 40%.

Short answer to your question - if the energy source is renewable then I'm comfortable with the

input:output equation. If not then ...

Would be really cool if each AI request - at the end, calculated the energy consumption used for that request and reported in some universal unit - there's the Maccas Burger 🍔 scale of relative costs. What about the burger scale of energy use? The level of energy used for this request was sufficient to cook 0.25 Burgers.

u/AutoModerator • 1 points • 2025-12-05 21:15 UTC

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If your post is a screenshot of a ChatGPT conversation, please reply to this message with the [conversation link](<https://help.openai.com/en/articles/7925741-chatgpt-shared-links-faq>) or prompt.

If your post is a DALL-E 3 image post, please reply with the prompt used to make this image.

Consider joining our [public discord server](<https://discord.gg/r-chatgpt-1050422060352024636>)! We have free bots with GPT-4 (with vision), image generators, and more!

🤖

Note: For any ChatGPT-related concerns, email support@openai.com

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